GAMELYTICS

Mobile Game Player Retention & Engagement Analysis

Product Analytics & Product Management Case Study

SQL • MySQL • Power BI • A/B Testing • Product Strategy

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1 EXECUTIVE SUMMARY

Core Question

How can player behavior data be used to identify patterns associated with sustained engagement, and which product intervention should be prioritized to improve player retention?

Project Objective

Analyze player authentication, retention and monetization behavior to identify engagement challenges, evaluate player segments and translate the findings into a measurable product recommendation.

Key Findings

- D1+ Rolling Retention 23.87%
- D7+ Rolling Retention 21.57%
- D30+ Rolling Retention 10.72%
- One-time players 76.16%
- 4.67% of players generate 79.32% of all authentication events
- Test Group A payer conversion 0.954%
- Test Group B payer conversion 0.891%

Core Product Insight

Player activity is highly polarized. 76.16% of registered players record only a single authentication event, while just 4.67% belong to the High-engagement segment and generate 79.32% of all authentication activity. Rolling retention declines from 23.87% at D1+ to 21.57% at D7+ and 10.72% at D30+, indicating that the largest deterioration occurs over the medium-to-long term. This pattern remains generally consistent across registration cohorts. This suggests a persistent engagement challenge rather than an issue focusing on a specific registration period. Higher authentication engagement also does not directly translate into proportionally stronger monetization, so sustained engagement and monetization should be treated as related but distinct product objectives.

Product Response

Problem: Low transition from initial interaction to sustained engagement

Hypothesis: Stronger return incentives may encourage sustained engagement

Prioritization: 3 feature proposals evaluated using RICE

Selected Proposal: Progressive Return Rewards

Validation: Event Tracking + Measurement Framework + A/B Test

DATA SCOPE: Registration, authentication, and A/B test revenue data. Authentication events are interpreted as observed activity signals, not gameplay sessions. Detailed data methodology, assumptions and limitations are covered in Data, Methodology & Analytical Scope.

2 DATA, METHODOLOGY & ANALYTICAL SCOPE

2.1 Data Source

Source: Kaggle-Gamelytics: Mobile Analytics Challenge (debs2x/gamelytics-mobile-analytics-challenge). The project uses this public mobile-analytics dataset as its raw analytical foundation. No synthetic gameplay events were added to fill gaps in the source data.

Why this dataset was selected: It provides player registration data, event authentication history, and user A/B-test revenue data, allowing retention, engagement, and monetization questions to be examined within one project while considering the data limitations.

2.2 Raw Data Structure

Source File	Records	Fields	Analytical Role
reg_data	1000000	reg_ts, uid	Registration anchor, retention eligibility and cohort assignment
auth_data	9601013	auth_ts, uid	Observed return activity, authentication frequency and observed lifespan
ab_test	404770	user_id, revenue, testgroup	Monetization metrics and existing A/B-group comparison

Table 1: reg_data is player-level registration data; auth_data is event-level and may contain many authentication records for the same uid; ab_test is user-level experiment/revenue data**.**

Completeness: No missing values were identified in the three imported source files.

Duplicates: No exact duplicate rows were identified in the imported source files.

Timestamp format: reg_ts and auth_ts are given in Unix timestamps. They were converted into calendar dates/datetimes in SQL whenever a date was required.

Database: The working database is gamelytics_project, with tables registrations, authentications, and ab_test.

2.3 Data Relationship & Analytical Layer

Table	Key Logic	Purpose
registrations	uid = player identifier; primary key	Lifecycle anchor for registration-based analysis
authentications	uid links to registrations.uid	Repeated observed activity after registration
ab_test	user_id links to player uid	Revenue and test-group outcomes

Table 2: Explanations of the data and the purpose in use.

The data is evaluated and translated through an analytical layer: SQL is used as the calculation layer for reproducible metrics and segmentation logic; Power BI is the visualization layer; the case study is the interpretation and product-decision layer. This prevents dashboard visuals from becoming the analytical logic itself.

2.4 Analytical Scope

Supported by the data set	Not supported by the data set
Rolling retention and registration cohorts	True gameplay-session duration or session frequency
Authentication-based engagement segmentation	Level progression, attempts, wins/failures or difficulty
Observed first-to-last authentication lifespan	Tutorial/level funnel analysis
Payer conversion, ARPU and ARPPU	Booster, lives, economy or feature-usage analysis
A/B-group conversion and revenue comparison	Acquisition-channel, device or geography analysis
Cross-analysis of matched engagement and revenue users	Direct behavioral root-cause diagnosis of churn

Table 3: The scope of data.

The final scope follows the fields that actually exist in the selected dataset. Analyses from the original project concept that require unavailable gameplay telemetry were excluded rather than artificially generated.

2.5 Operational Metric Definitions

Metrics	**Definitions in this project**
D1+ / D7+ / D30+ Rolling Retention	Among players with sufficient observable time to reach the threshold, the share with at least one authentication event on or after day 1 / 7 / 30 following registration.
Authentication Count	Total number of recorded authentication events for a player. Used as an activity-frequency index, it is not a gameplay-session count.
Observed Lifespan	Day difference between a player's first and last recorded authentication event within the dataset.
Payer	A user with recorded revenue greater than 0.
Payer Conversion	Paying users divided by all users in the relevant analytical players.
ARPU	Total recorded revenue divided by all users in the relevant players.
ARPPU	Total recorded revenue divided by paying users in the relevant players.

Table 4: Explanations of the metrics used.

These definitions establish the calculation rules used later in the analysis. Results and product interpretation are intentionally presented in the following relevant sections.

2.6 Retention & Cohort Methodology

Lifecycle evaluation: Registration timestamp (reg_ts) is the starting point for retention eligibility and cohort analysis.

Observation of endpoint: The latest observed authentication date is 23 September 2020. A player is included in a D1+, D7+, or D30+ denominator only when the available observation window is long enough for that threshold to be part of the denominator.

Rolling retention rather than exact-day retention: The primary retention framework asks whether a player was observed again on or after the threshold. This is better aligned with the available authentication-event structure and avoids treating exact-day returns as a static retention curve.

Cohort definition: Registration year is used for the cohort comparison. The main visualization focuses on 2012–2019. 2020 is not included as it is incomplete for the analysis because observations end during that year.

Interpretation boundary: The dataset contains no campaign, release, acquisition channel, or change in product that could explain cohort differences respectively.

2.7 Engagement Segment Methodology

Segment	Rule	Interpretation
One-time	1 authentication	Observed only once in authentication history
Low	2-5 authentications	Limited repeated authentication activity
Medium	6-11 authentications	Moderate repeated authentication activity
High	12+ authentications	Highest-frequency authentication segment

Table 5: Generated segments and their references.

Engagement segments are defined regarding total authentication-event count per player. The thresholds were selected from the observed frequency distribution, including a significant break between 11 and 12 authentication events. These are analytical frequency segments, not predefined player personas. These segments only define observed authentication behavior and do not explain motivation, satisfaction or churn cause.

2.8 Monetization Matching Methodology

Cross-dataset coverage: 364,555 of the 404,770 A/B-test users can be linked to the authentication population through the player id, leaving 40,215 unmatched users. The exact match is 90.0647%.

The scope: The analyses that compare engagement segments with monetization use only this matched population. The unmatched users are not assigned to an engagement segment, and matched-population findings are not generalized to all registrations.

2.9 Explicit Assumptions

Authentication as an activity signal: auth_ts is evidence that a user was observed authenticating. It does not represent a complete gameplay session or meaningful in-game action.

Registration as the lifecycle start: reg_ts is used as the starting point of user activity for retention calculations because it is the available player entry timestamp.

Revenue $>$ 0 defines a payer: The dataset does not provide any event-based purchase detail, so payer status is decided regarding the user based revenue field.

Observed lifespan does not play a critical role: Based on the data, first-to-last authentication measures only the period visible in the dataset, not the player's true lifetime in the product.

Segmentation is limited: Authentication-frequency thresholds are generated to observe player behavior for comparison, they do not create or represent causal player states.

2.10 Data Limitations & Analytical Guardrails

No gameplay data: The source does not support claims about level difficulty, progression friction, tutorial completion, session duration, booster usage, economy behavior or feature interaction.

No direct evidence for churn: The analysis can identify where sustained activity is weak and which patterns coexist with retention or monetization outcomes. However, it cannot state why a player stops returning.

Limited observation window: Later registrations cause less observation time. Regarding eligibility, the 2020 cohort is not treated as a fully observed annual cohort.

Lacking monetization match: Engagement–monetization analysis is restricted to the 364,555 matched A/B users.

Unknown test context: The given A/B results in the dataset can support statistical comparison of observed outcomes, but not a causal product context about a specific feature because treatment metadata is absent.

Association is not causation: Observational findings are used to identify product opportunities and create hypotheses. They are not presented as proof of any cause in the product activity.

Future product hypotheses require new elements: Any proposed feature should be evaluated with its own event tracking plan, KPI framework, guardrails, and controlled experiment. Those elements are developed later in the case study rather than pre-judged in this methodology section.

3 RETENTION & COHORT ANALYSIS

Business Question 1

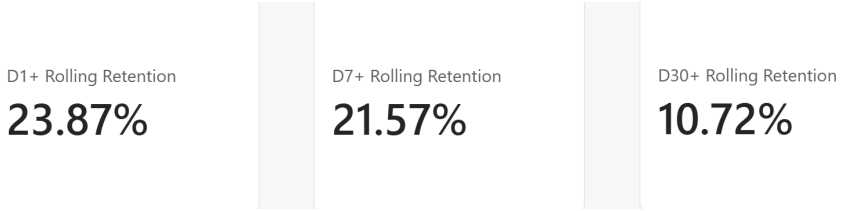
How does player retention change over time, and is the long-term retention challenge concentrated in specific registration cohorts or persistent across the player base?

3.1 Rolling Retention Performance

Metric	Retained players	Eligible players	Rolling retention
D1+	238378	998747	23.87%
D7+	213375	988942	21.57%
D30+	102099	952238	10.72%

Table 6: Rolling retention across eligible players.

Rolling Retention rates are given in Visual 1. The share of eligible registered players with an authentication event on or after day 30 is 10.72%, compared with 23.87% at day 1 and 21.57% at day 7. The D7+–D30+ rate difference is 10.85 percentage points.

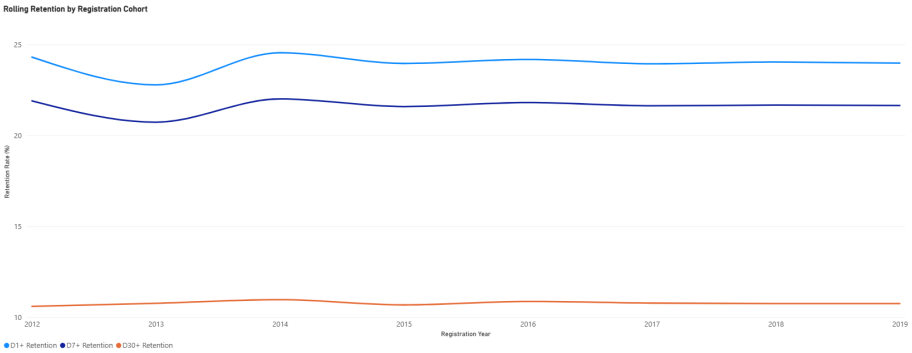


Visual 1: D1+, D7+, and D30+ Rolling Retention rates.

The biggest difference in the reported rolling retention rates appears between the day-7 and day-30 thresholds. This is consistent with a long-term engagement challenge, as comparatively fewer eligible users are observed authenticating at or beyond day 30. Since eligibility denominators differ across thresholds, the 10.85 percentage point difference is not itself a player churn probability.

3.2 Retention by Registration Cohort

The cohort demonstration in Table 7 below reproduces the retention rates underlying the completed Power BI dashboard. The values given in Table 7 are presented so that the analytical evidence remains legible outside the Visual 2.



Visual 2: Rolling Retention by Registration Cohort.

Registration year	**D1+**	**D7+**	**D30+**
2012	24.31%	21.90%	10.60%
2013	22.78%	20.73%	10.77%
2014	24.55%	22.01%	10.97%
2015	23.96%	21.59%	10.68%
2016	24.18%	21.81%	10.87%
2017	23.94%	21.63%	10.78%
2018	24.04%	21.67%	10.75%
2019	23.98%	21.65%	10.75%

Table 7: Rolling retention across registration year cohorts.

Across the 2012–2019 registration cohorts, D30+ retention ranges from 10.60% in 2012 to 10.97% in 2014, a spread of 0.37 percentage points. D1+ and D7+ vary more, including a lower result for the 2013 cohort, but no cohort shows a notably different D30+ outcome.

The consistency of D30+ across registration years suggests that lower observed activity for long-term is a recurring pattern in this dataset rather than a problem in one registration cohort. The cohort comparison does not state causes, such that no campaign, game-release, onboarding-change or player-acquisition metadata is available to explain the differences.

3.3 Product Implications

The analysis supports prioritizing sustained return behavior, rather than focusing solely on the first post-registration authentication.

The similar D30+ pattern across cohorts suggests evaluating larger applicable retention interventions before targeting a single registration year.

The proposed intervention should be assessed with a defined retention KPI and

instrumentation that distinguishes actual gameplay engagement from authentication events.

3.4 Decision Boundary

These results show where the retention opportunity is observed, not why it occurs. The feature hypotheses, prioritization, and future experimental design are mentioned in the later Product Management sections. This section does not claim that any particular reward, progression, or notification mechanic caused the observed retention pattern.

4 PLAYER ENGAGEMENT ANALYSIS

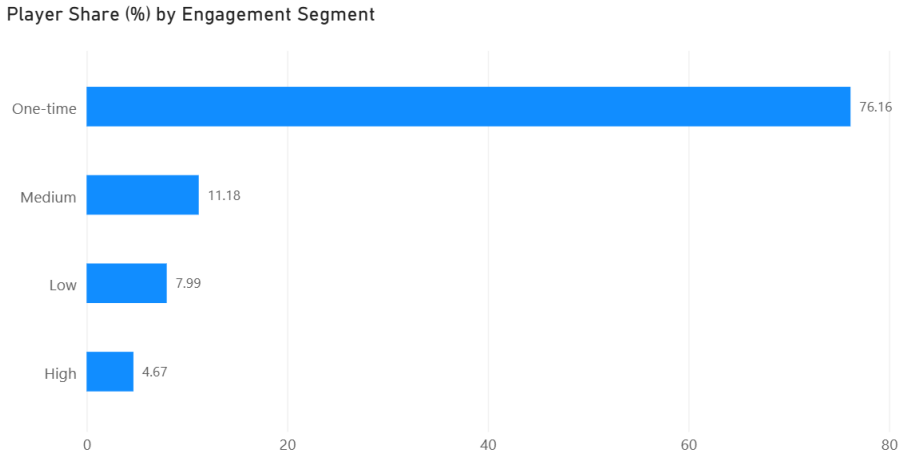
Business Question 2

How is observed authentication activity distributed across players, and what does this reveal about the concentration and persistence of engagement?

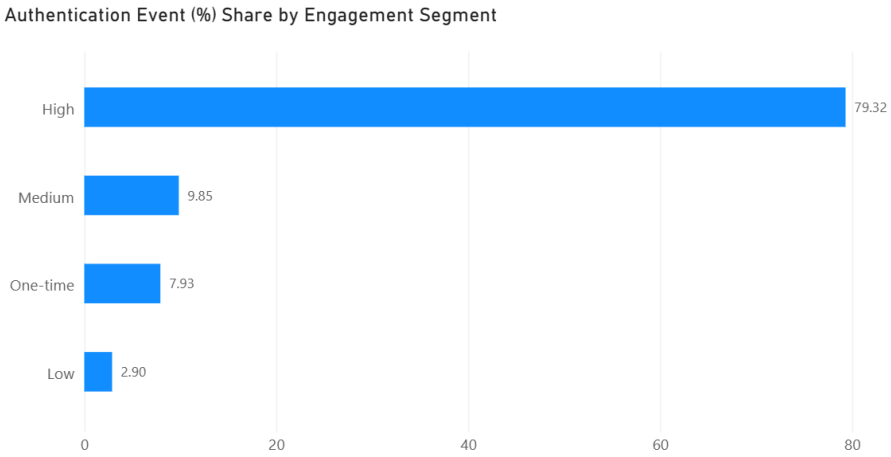
4.1 Key Findings

Observed activity is highly concentrated. Of 1,000,000 players, 761,622 (76.16%) have exactly one recorded authentication event. Conversely, 46,670 players (4.67%) generate 7,615,467 of 9,601,013 authentication events (79.32%). The dataset establishes this distribution, but not its behavioral cause.

4.2 Engagement Segment Performance



Visual 3: Player Share (%) by Engagement Segment



Visual 4: Authentication Event (%) Share by Engagement Segment.

The engagement distribution demonstrates a notable imbalance between the number of players in each segment and their contribution to total authentication activity.

One-time players account for 76.16% of the registered players but generate only 7.93% of recorded authentication events, Visual 4. On the other hand, the High engagement segment accounts for 4.67% of players while contributing 79.32% of all authentication activity, Visual 3. This contrast shows that the size of a player segment is not necessarily proportional to its contribution to overall authentication activity, considered as product activity.

The observed distribution suggests that aggregate authentication activity is considerably responsive to the High engagement players. As a result, an increase in total authentication events could occur even if the majority of players do not become more engaged. A decline in the activity among High players could reduce the total event volume without a comparable change in the number of registered players. Total authentication events in isolation are not evidence of engagement improvement. Activity volume and engagement can also be evaluated through players with more than one authentication, distribution of authentication frequency, and retention performance.

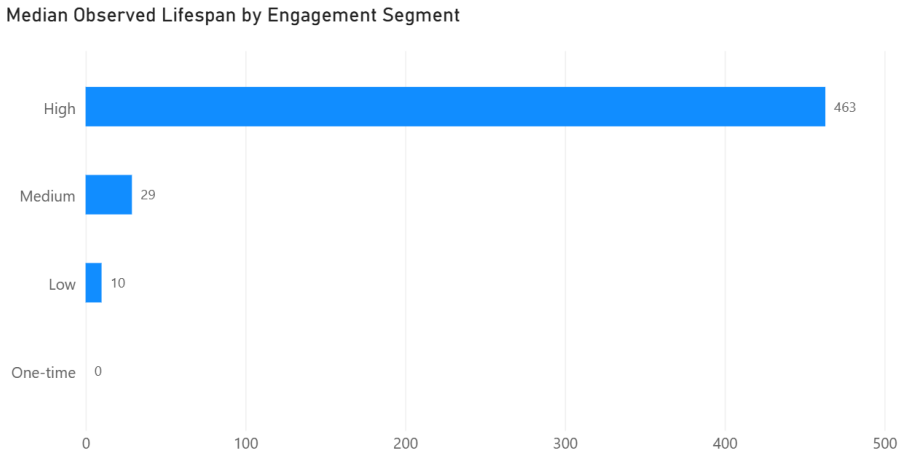
Product Management Implication

The product team should differentiate between two objectives: increasing engagement depth among already active players, and increasing engagement breadth by encouraging more players to return. The current analysis demonstrates the relevance of both objectives, but does not state which objective generates more business value.

4.3 Observed Player Lifespan

Authentication frequency provides information about how often players interact with the product, while observed lifespan captures the time elapsed between their first and last recorded authentication events.

Analyzing these dimensions together helps distinguish players who generate repeated activity within a limited observation window from those whose recorded activity extends over a longer period.



Visual 5: Median Observed Lifespan by Engagement Segment.

The median observed lifespan increases substantially across engagement segments, reaching 463 days for High players compared with 29 days for Medium players, Visual 5.

This indicates that players with more recorded authentication events also tend to have longer observed activity windows. However, this relationship requires careful interpretation.

Frequency–Lifespan Relationship

Authentication frequency and observed lifespan are related but distinct measurements.

A player may authenticate repeatedly within a short period, while another may authenticate less frequently over several months. Both behaviors contribute to engagement, but they represent different activity patterns. The available segment data does not allow these patterns to be fully distinguished.

Furthermore, players with longer observed lifespans have more opportunities to accumulate authentication events. Consequently, the relationship between authentication frequency and lifespan should not be interpreted as evidence that higher authentication frequency independently causes longer-term engagement.

Mean vs. Median Interpretation

The High segment has a mean observed lifespan of 647.90 days and a median of 463 days. The mean exceeding the median suggests that some particularly long observation windows increase the average lifespan of this segment.

This makes the median more analytical for describing the tendency of observed lifespan in the High segment, although a complete assessment of distributional difference would require additional player data.

Analytical Limitation

Observed lifespan is not equivalent to true customer lifetime. It measures the interval between recorded authentication events rather than the time until permanent player churn. A player whose final authentication appears in the dataset may return after the observation period ends.

Therefore, the reported values should be interpreted as observed activity windows rather than confirmed player lifetimes.

Product Management Implication

The combined frequency and lifespan analysis suggests that engagement quality should not be assessed through event frequency alone.

Future product measurement should distinguish between:

- Frequency of return activity.
- Duration of observed engagement.
- Consistency of return behavior across time.

These dimensions would help to determine whether an intervention encourages temporary fluctuations of activity or more sustained player engagement.

4.4 Segment Level Product Opportunities and Measurement Risks

The engagement segmentation identifies different patterns of observed activity, but each segment presents a different analytical challenge.

One-time Players: Limited Evidence of Repeated Activity

The One-time segment contains 761,622 players, representing 76.16% of the registered player base. Its size indicates that repeated authentication is not observed for most players in the dataset. However, a single recorded authentication does not necessarily mean that the player abandoned the game immediately. The dataset does not reveal gameplay duration, session quality, or activity outside the recorded observation period.

The relevant product question is therefore not why these players churned, but what factors may be associated with their lack of observed repeat authentication. Potential areas for future investigation include early progression, initial difficulty, and exposure to return incentives. These are proposed investigation areas rather than explanations given by the current data.

Low and Medium Players: Intermediate Activity Patterns

Together, the Low and Medium segments account for 191,708 players, or approximately 19.17% of the player base.

These players demonstrate repeated authentication but remain below the High segment's activity threshold.

Their behavior may be especially useful for future player analysis because they provide evidence of repeated interaction without the exceptionally high event counts observed in the High segment.

However, the current aggregate data does not state that Low players naturally progress into Medium or High engagement segments. The segments describe observed cumulative activity, not sequential stages in a player journey.

An analysis would be required to determine whether players transition between engagement states over time.

High Players: Concentrated Activity and Potential Measurement Distortion

High players represent only 4.67% of the population but contribute 79.32% of recorded authentication events. This makes the segment particularly influential in aggregate activity reporting.

For example, a feature that increases authentication frequency among existing High players could substantially increase total authentication volume without expanding the number of players who return.

As a result, product success should not be measured exclusively through aggregate authentication growth. The High segment also should not automatically be classified as the most valuable customer segment. Authentication frequency is not equivalent to revenue contribution, and this relationship requires a separate monetization analysis.

4.5 Product Implications and Measurement Priorities

Priority 1 - Improve Engagement Breadth

The large One-time segment motivates the analysis to offer opportunities to encourage repeated authentication among players with limited observed activity.

A relevant objective would be to increase the proportion of eligible players who demonstrate repeat activity, rather than focusing on increasing total authentication volume.

Priority 2 - Measure Engagement Depth Separately

The concentration of activity among High players indicates that engagement depth should be monitored independently from engagement breadth.

An increase in average authentication frequency may be driven by a small number of highly active players and should not automatically be interpreted as improvement across the broader player population.

Priority 3 - Distinguish Sustained Engagement from Activity Spikes

Authentication frequency should be evaluated together with time-based measures such as rolling retention and observed activity duration.

This would help distinguish sustained return behavior from short periods of intensive activity.

Priority 4 - Validate Product Hypotheses Before Implementation

The observed engagement distribution identifies potential opportunities but does not directly state their causes.

Product interventions involving progression, rewards, or re-engagement should be treated as hypotheses that require further behavioral instrumentation and controlled experimentation.

5 A/B TEST & MONETIZATION ANALYSIS

Business Question 3

Do players with more observed authentication activity demonstrate stronger payer conversion or revenue performance, and what does this imply for product prioritization?

5.1 Key Findings

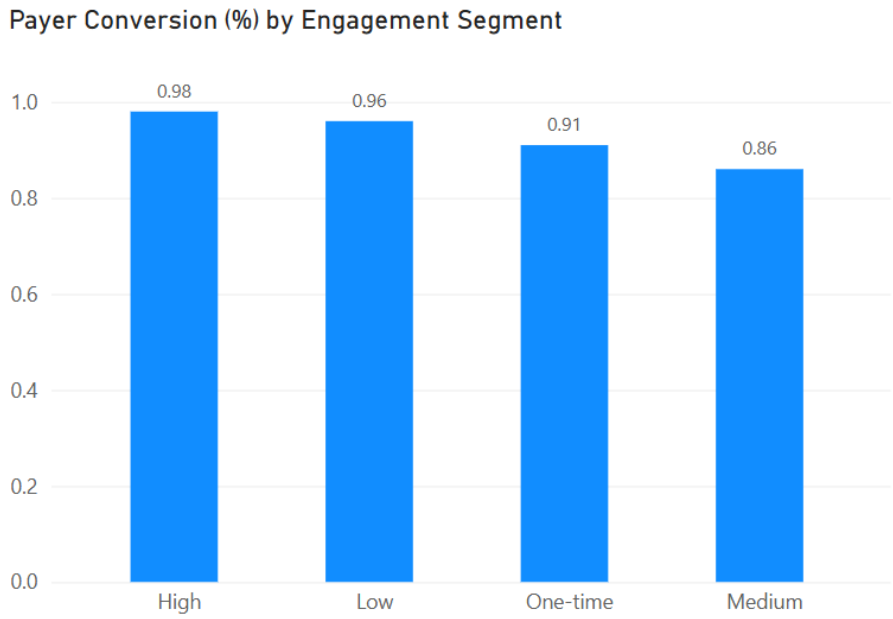
Higher authentication frequency does not correspond to consistently stronger monetization outcomes in the matched dataset. Payer conversion ranges from 0.86% to 0.98% across engagement segments, while ARPU ranges from 23.52 to 28.15 revenue units. The High engagement segment generates more authentication events, but it does not lead the matched segments in ARPU.

5.2 Matched Engagement Revenue Population

The engagement monetization comparison uses 364,555 users matched between the authentication-based segmentation and the A/B revenue table, representing approximately 90.06% of the 404,770 users in the revenue dataset. The remaining 40,215 users are not included in segment comparisons. This is a matched subset analysis, not a complete monetization profile of all registered players or all revenue users.

ARPU and ARPPU use the revenue field's recorded units; the source does not establish a currency. Displayed percentages and averages are rounded. Payer conversion is the share of matched users with positive recorded revenue.

5.3 Payer Conversion



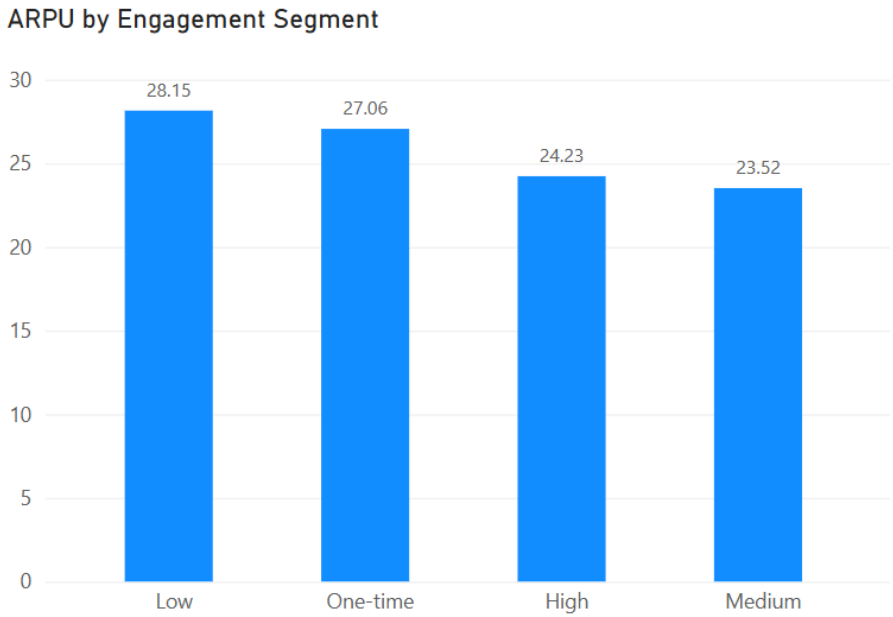
Visual 6: Payer Conversion (%) by Engagement Segment.

The High segment has the highest observed payer conversion of 0.98%, followed by Low 0.96%, One-time 0.91%, and Medium 0.86%, Visual 6. However, the overall difference is only 0.12 percentage points. Conversion does not rise steadily as authentication frequency increases. The descriptive differences are insufficient to claim that as engagement intensity increases, it is more likely for players to make payment.

The practical implication is that the volume of recorded authentication events should not be used as a standalone payer activity indicator. Conversion outcomes need to be evaluated directly, with uncertainty and segment composition

considered before targeting a particular group.

5.4 Average Revenue per User (ARPU)



Visual 7: ARPU by Engagement Segment.

The Low segment records the highest ARPU of 28.15, while Medium records the lowest 23.52. High ARPU is 24.23, below One-time ARPU of 27.06, Visual 7. This ordering challenges the assumption that the most frequently authenticating players necessarily deliver the highest recorded revenue per user. It does not imply that less frequent authentication causes greater revenue.

5.5 Measurement Risks

The 40,215 revenue users without a corresponding authentication-based segment may differ systematically from matched users. Without comparing their revenue distributions and characteristics, results cannot safely be generalized to all A/B users. Additionally, engagement segments based on cumulative authentication counts suggest that longer observed users may have more opportunity to enter higher frequency categories. This makes cross-segment monetization comparisons descriptive rather than causal.

Recorded revenue is highly zero-inflated, as payer conversion is below 1% in every segment. Consequently, ARPU combines two mechanisms: whether a user pays and how much a payer spends. ARPPU is sensitive to the composition of a small paying population.

5.6 Product Implications

A feature that raises authentication activity should not be assumed to increase revenue. Evaluate retention regarding experiments with monetization guardrails such as payer conversion and ARPU, rather than making revenue increase as the only measure of success. Before designing segment-specific monetization features, test whether distinct differences persist after controlling for observation window and registration cohort. Avoid describing the High segment as the most valuable segment without evidence on incremental revenue, costs and player lifetime value.

Business Question 4

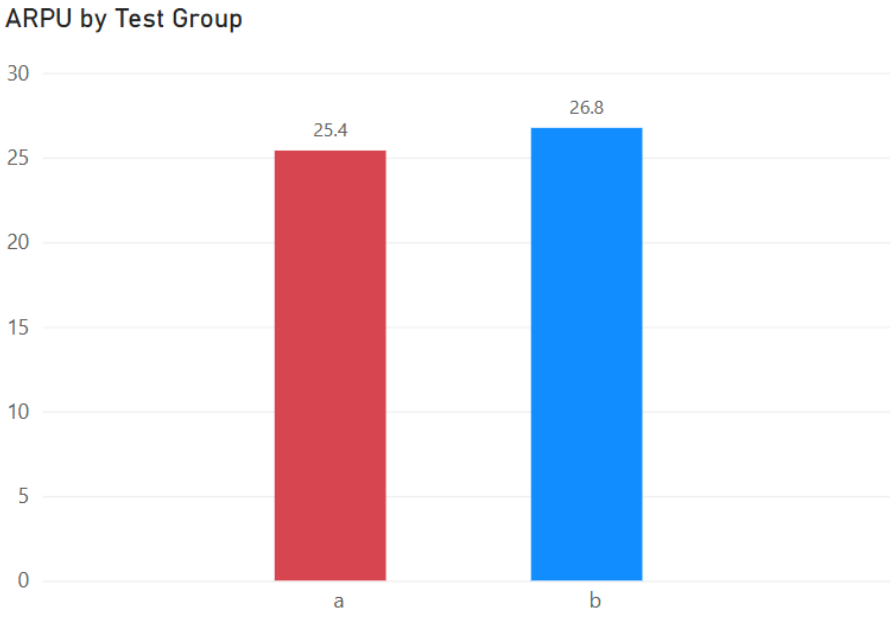
Does either test group demonstrate a statistically supported advantage in payer conversion or revenue performance, and is the evidence sufficient to recommend a winning variant?

5.7 Key Findings

Group A has a higher observed payer conversion rate of 0.95% versus 0.89% for Group B, Visual 8. A two-sided pooled two-proportion z-test indicates evidence of a difference in conversion, p 0.035. In contrast, Group B has higher ARPU 26.75 versus 25.41 revenue units, Visual 9, and ARPPU 3004 versus 2664 revenue units, Visual 10. These metrics point in different directions. Without a defined primary success metric, experiment context, and a statistical comparison of revenue outcomes, the analysis does not come up with an overall winner among the test groups.



Visual 8: Payer Conversion (%) by Test Group.



Visual 9: ARPU by Test Group.



Visual 10: ARPPU by Test Group.

5.8 Test Performance

Metric	Group A	Group B
Users	202103	202667
Payers	1928	1805
Payer Conversion	0.95%	0.89%
ARPU	25.41	26.75
ARPPU	2664	3004
Payer Revenue Range	200-37433	2000-4000

Table 8: A/B Test evaluations by test groups.

Revenue values are presented in dataset units, the currency is not given. Payer conversion is the proportion of users with positive revenue. ARPU is total revenue divided by all assigned users, ARPPU is total revenue divided by paying users. The reported rounded revenue metrics should not be used for the exact transaction distributions.

5.9 Statistical Evidence

The observed absolute difference between A and B is 0.0633 percentage points, equivalent to approximately 7.1% higher conversion relative to Group B. Under

a two-sided pooled two-proportion z-test of equal conversion rates, z = 2.11 and p = 0.035. An approximate unpooled 95% confidence interval for A and B is [0.0044, 0.1222] percentage points.

At the most commonly used 5% significance threshold, the observed conversion difference is statistically significant under the test assumptions. Statistical significance does not suggest that the effect is economically notable or reflective of unverified test issues. In particular, the dataset does not establish randomization quality, pre-specified hypotheses, testing duration, sequential monitoring, or whether conversion was the experiment's primary endpoint.

5.10 Revenue Metrics Complications and Decision Boundaries

Although fewer Group B users become payers, Group B generates higher observed revenue per user and revenue per payer. This is mathematically possible because payer conversion and conditional spending describe different components of monetization. The pattern suggests a trade-off between breadth of monetization, how many users pay, and intensity among payers, how much payers spend, but it does not explain why the difference occurs.

Group A includes the highest value of 37433, whereas the reported Group B payer range is 2000–4000. This causes a material range difference. These ranges signal that the distributions may have different shapes and outliers. They do not specify variance, distributional significance, data errors, or the impact of any single observation. A robust comparison of ARPU would require user revenue data, including zeros, and an appropriate analytical method such as a randomization-based test.

The conversion test supports a narrower statement that payer conversion differs between the two recorded groups under the stated assumptions. It does not demonstrate that Group A is the better overall product variant. On the other hand, Group B's higher descriptive ARPU does not establish a statistically significant revenue advantage. The treatment and control mapping, intervention details, target population, product objective, and cost implications are not available. A causal explanation or a recommendation would therefore be premature and inappropriate at the moment.

A/B test user assignment balance is close in count 202103 versus 202667, respectively, but similar sample sizes alone do not prove successful randomization. A formal sample ratio mismatch assessment requires the intended and same allocation ratio, which is not documented here.

5.11 Product Management Implications

The decision principle is to select and pre-register a primary KPI before evaluating a new experiment. If payer conversion is primary, Group A shows a statistically supported observed advantage, but if ARPU is primary, the available summary is

insufficient to determine significance. A decision should also consider guardrails, practical effect size, user experience, and experiment validity.

The analytical next step is to obtain user revenue observations and experiment metadata, verify allocation and exposure logic, quantify uncertainty around ARPU, test zero inflation and upper-tail sensitivity, and assess whether the effect is stable across relevant segments without misleading multiple comparisons.

The portfolio takeaway is that the responsible conclusion is not "A wins" or "B wins," but that conversion and revenue outcomes differ, and a valid product decision requires an explicitly defined objective and appropriate statistical evidence.

6 PRODUCT PROBLEM & HYPOTHESIS

Decision Question

Given the observed retention and engagement patterns, what product problem is to prioritize, what remains unknown, and which hypothesis should guide the next stage of solution exploration?

6.1 Evidence to Problem Synthesis

The analysis covers specific observable challenges that the repeat activity is not broadly distributed, and sustained return activity is limited. The evidence states that the analysis does not directly identify the underlying behavioral cause.

Evidence base	**Analysis establishment**	**Not established**
Retention	D30+ rolling retention is 10.72%; the long-horizon return signal is limited.	Why a player stops returning or which feature would change that behavior.
Cohorts	D30+ varies only from 10.60% to 10.97% across 2012–2019 cohorts.	That all cohorts have identical player composition or acquisition conditions.
Engagement	76.16% have one recorded authentication; 4.67% generate 79.32% of authentication events.	That one authentication equals one gameplay session or confirms immediate churn.
Monetization	Higher authentication frequency does not consistently correspond to higher ARPU.	That greater engagement has no business value or cannot improve revenue.

Table 9: What the analysis establishes and what does not on the evidence base.

These are descriptive findings, but not causal. The product problem should be in the scope of the observable outcome and limited sustained return behavior, rather than an assumed mechanism such as inadequate rewards or poor onboarding of users.

6.2 Product Problem Statement

A large proportion of registered players show limited repeat authentication, while long-term return activity remains low across registration cohorts. The product opportunity is to increase sustained and broadly distributed return behavior without only relying on activity growth from a small high authentication frequency segment.

This statement avoids treating authentication events as gameplay sessions, considering users authenticated once as churners, and coming up with outcomes for any unobserved feature.

6.3 Desired Behavioral Change and Root-Cause Uncertainty

Dimension	**Working Definition**
Initial target	Eligible registered players with limited observed repeat authentication, eligibility and recent activity must be defined at experiment time.
Current behavior	No or limited recorded authentication during a specified observed window.
Desired behavior	A higher share of players returns and authenticates again, with improvement sustaining to long term retention.
Product outcome	Improvement in D30+ retention, accompanied by healthy D1+/D7+ trends and no unacceptable monetization or experience deterioration.
Scope restriction	The dataset does not identify onboarding completion, progression, reward exposure, motivations, or session quality.

Table 10: Target population and behavioral expectancies.

The one time segment can not be used directly as a test eligibility rule because segment assignments depend on future observations of product and user activity.

This is why a prospective test must define eligibility using only information given and available at the assignment.

The available data supports several acceptable mechanisms but cannot distinguish among them. Players may lack a reason to return, encounter problems during early experiences, face progression barriers, or simply have different preferences and usage opportunities. Observation-window differences can also affect activity classification. However, these explanations in Table 11 should not be reported as a finding because of the lacking information.

Hypothesis base	**Additional Evidence Required**
Insufficient return motivation	Feature exposure, return incentive eligibility, reward claims, time to next return.
Early experience problems	Tutorial completion, level failure and abandonment events.
Progression difficulty	Level attempts, failure rates, progression speed and repeated blockers.
Player distribution	Acquisition channel, registration cohort, platform, country and comparable observation windows.

Table 11: Product hypothesis regarding lacking information.

The primary product hypothesis: If eligible players receive a clearly communicated, progressive incentive for returning over multiple days, then a greater proportion will demonstrate sustained return authentication, measured primarily through D30+ retention, compared with players who do not receive the intervention.

The rationale behind the hypothesis is that a structured return incentive is one acceptable mechanism for encouraging repeated interactions. Its selection as a candidate is an informed product hypothesis, not a conclusion derived from feature evidence.

6.4 Assumptions, Risks and Product Decision

Category	**Testable Consideration**
Behavioral assumption	Players notice, understand and value the proposed return incentive.
Mechanism assumption	The incentive changes return decisions rather than merely shifting authentication timing.

Measurement risk	An increase in authentication events may be concentrated among already-active users.
Business risk	Incremental engagement may not translate into revenue or may increase incentive costs.
Counter evidence	No meaningful D30+ improvement, only short-lived early lifts, or adverse guardrail changes.

Table 12: Regarding considerations based on assumptions and risks.

The hypothesis should be rejected or revised if a sufficiently powered and correctly implemented experiment does not show a practically meaningful improvement. Referring to the assumptions and risks considered in Table 12, an observed increase in total authentication volume, by itself, would not validate the hypothesis.

The evidence justifies investigating interventions designed to enlarge sustained return activity. It does not yet justify implementation of a specific feature. Solution alternatives will be compared by using explicit reach, impact, confidence and effort assumptions. Additionally, the selected concept will be translated into event instrumentation, metric definitions and an experiment plan.

It is decided to prioritize the sustained return problem, keep the proposed progressive incentive as a candidate hypothesis, and temporarily suspend feature commitment until alternatives and measurement feasibility are evaluated.

7 FUTURE ALTERNATIVES & RICE PRIORITIZATION

Business Question 5

Which product initiative should be prioritized to improve sustained player engagement, considering its expected reach, potential impact, confidence and implementation effort?

7.1 Proposed Future Alternatives

Alternative 1 - Progressive Return Rewards:

A structured reward system that offers increasingly valuable incentives for returning across multiple days. A visible progression of rewards may encourage players to return repeatedly rather than engaging only once.

Alternative 2 - 7 to 30 Day Mission:

A sequence of short- and medium-term objectives that encourages players to complete recurring challenges. Clear progression milestones may provide continued motivation and strengthen longer term engagement.

Alternative 3 - Personalized Reengagement:

Targeted return notifications or offers based on a player's previously observed behavior and inactivity. Relevant communication may encourage inactive players to return by reducing the gap between the product experience and individual preferences.

7.2 Prioritization

The alternatives are evaluated through the RICE prioritization framework.

Reach: Estimated relative number of players who could be affected. The value range is 0-3.

Impact: Expected influence on the sustained engagement objective. The value range is 0-3.

Confidence: Confidence in the underlying assumptions. The value range is in percentages.

Effort: Relative implementation effort. The value range is 0-3.

The values for the completion of the RICE framework are assigned through the assumptions and risk. The value ranges are also considered accordingly.

Feature	**Reach**	**Impact**	**Confidence**	**Effort**	**RICE**
Progressive Return Rewards	3	2	80%	1	4.80
7–30 Day Progression Missions	3	3	80%	3	2.40
Personalized Re-engagement	2	2	50%	2	1.00

Table 13: Feature prioritization results in the RICE framework.

7.3 Analytical Interpretation

The ranking in Table 13 depends on subjective planning estimates, this is why it should not be treated as a definitive prediction of feature performance. The framework and estimates show that the preferred alternative is sensitive to implementation complexity. The initial prioritization should be revisited after engineering estimates, product design requirements and reward economy constraints.

Progressive Return Rewards achieves the highest RICE score of 4.80, having the highest priority. Its main advantage is the combination of potentially broad reach and relatively low implementation effort. A simple reward progression could be introduced without requiring extensive personalization or complex mission systems. The feature also provides a clearly defined behavioral mechanism in which players receive an additional reason to return on subsequent days. However, the dataset does not demonstrate that insufficient rewards cause low retention. Therefore, the intervention is best understood as a testable hypothesis rather than a proven solution.

7-30 Day Progression Missions receive a RICE score of 2.40. Although this alternative has the highest assumed impact, it requires a higher implementation effort. The feature may create more meaningful engagement through structured and consecutive objectives, but it would require additional design, balancing, progression tracking and supporting event instrumentation. Its lower RICE score does not imply that it would be less effective in practice. It reflects the current assumptions about relative cost and potential benefit.

Personalized Re-engagement receives a RICE score of 1.00. Its lower score is driven by narrower assumed reach, greater implementation requirements than basic rewards, and limited confidence in the available data. The current dataset does not contain metadata including player preferences, gameplay progression or feature interaction histories. Consequently, meaningful user based product orientation would require additional behavioral instrumentation before this approach could be evaluated reliably.

7.4 Product Implication

Based on the RICE evaluation, Progressive Return Rewards is selected as the preferred candidate for further experimentation.

The recommendation is based on three considerations:

1. It directly targets repeated return behavior, which is aligned with the product problem.
2. It offers a relatively simple mechanism that can be tested before investing in more complex engagement systems.
3. Its estimated reach by effort ratio is stronger than the alternatives under the current assumptions.

The recommendation is to create a prototype and test the feature, not to proceed immediately. The success must be established through observed changes in sustained retention rather than increases in reward claims or authentication volume alone.

The measurable experiments will determine whether Progressive Return Rewards produces a meaningful improvement in sustained player retention while maintaining acceptable monetization and player experience outcomes.

8 MEASUREMENT FRAMEWORK & EXPERIMENT DESIGN

Business Question 6

How should Progressive Return Rewards be measured and experimentally evaluated to determine whether it improves sustained player retention without negatively affecting monetization or player experience?

8.1 KPI Framework

Primary KPI:

D30+ Rolling Retention

The proportion of eligible randomized players with at least one recorded authentication on or after Day 30 within a prespecified observation scope.

For the proposed experiment design, a 60 day post assignment observation window is generated as a planning assumption. To create the experiment, this assumption states that players qualify for D30+ retention if at least one authentication occurs between Day 30 and Day 60 after assignment.

Supporting KPI's:

D1+ Retention

D7+ Retention

Repeat Authentication Rate

Authentication Frequency

Reward Claim Rate

Reward Progression Completion

Guardrail KPI's:

ARPU

Payer Conversion

Reward Economy Cost

Game Progression Balance

8.2 Event Tracking Plan

Event	Properties	Measurement Purpose
reward_offer_viewed	user_id, reward_day, timestamp	Measure feature exposure
reward_claimed	user_id, reward_day, reward_type, reward_value, timestamp	Measure reward utilization
reward_streak_completed	user_id, streak_length, timestamp	Measure reward progression
authentication	user_id, timestamp	Measure return behavior
session_start	user_id, timestamp, session_id	Measure gameplay session initiation
session_end	user_id, timestamp, session_id, duration	Measure gameplay session duration
level_start	user_id, level_id, timestamp	Monitor progression activity
level_complete	user_id, level_id, attempts, timestamp	Monitor progression outcomes
purchase_completed	user_id, transaction_id, amount, currency, timestamp	Measure monetization
reward_currency_spent	user_id, currency_type, amount, timestamp	Monitor reward economy effects

Table 14: Required events and measurement purposes for experiment design.

It is seen that authentication and gameplay sessions are set apart as separate event concepts. Table 14 shows what events are required to generate the tracking plan of the experiment. All events should use consistent user identifiers and timestamp to analyze the events under a standardized experimentation.

8.3 Future A/B Test Plan

To create the experimental configurations, the experiment evaluates whether Progression Return Rewards causes an improvement in sustained return behavior.

Players with existing experience are assigned in Control Group A, who are eligible players without the Progressive Return Rewards feature.

Players with Progressive Return Rewards feature are assigned in Treatment Group B, who are eligible to receive the proposed multi-day reward experience.

Component	**Approach**
Randomization	User based random assignment
Allocation	50% Group A / 50% Group B
Control	Existing product feature and experience
Treatment	Progressive Return Rewards
Primary KPI	D30+ retention within a fixed 60-day observation scope
Statistical significance	Two-sided $= 0.05$
Statistical power	80% targeted
Target Retention Improvement	+1.50 percentage points

Table 15: A/B test components and their proposed approach.

The historical A/B groups analyzed previously are unrelated to this proposed experiment and approach as in Table 15. The original treatment definitions in the dataset are unknown and should not be used to infer the effect of Progressive Return Rewards experiment.

8.4 Measurement Risk and Limitations

Selection and assignment bias: Eligibility must be defined before treatment exposure, and assignment integrity should be checked. Players should not be excluded from the primary analysis based on reward feature usage.

Novelty effects: Short-term increases in activity may not persist a sustained reward experience. This reinforces the importance of long term retention measurement.

Metric deterioration: Repeated authentication events may increase without an improvement in gameplay engagement or player satisfaction. Therefore, authentication metrics should be complemented by session and progression events.

Economic trade-offs: Reward incentives may increase activity while reducing purchase demand. Monetization, reward, and cost guardrails are important.

8.5 Product Implication

This experiment should test whether Progressive Return Rewards produces a meaningful improvement in D30+ retention without distorting monetization performance and gameplay experience.

The most critical point is to distinguish feature engagement from product success. Reward claims and increased authentication volume may indicate that players

interact with the feature, but the product hypothesis would be supported only if sustained return behavior improves under the experimental design.

9 FINAL RECOMMENDATIONS

9.1 Executive Decision

The Gamelytics data analysis identifies sustained player engagement as the primary product improvement opportunity. The data analysis indicates that the recorded authentication activity is mostly concentrated among a small proportion of players, while longer term retention remains limited across registration cohorts.

Based on the combined analytical findings and structured feature prioritization, the recommended next step is to create a prototype and experimentally evaluate Progressive Return Rewards feature. The recommendation represents a decision to validate a promising product opportunity. It does not suggest that the proposed feature has already demonstrated effectiveness.

The only product objective is not improving aggregate activity volume. Therefore, the recommended strategy emphasizes engagement breadth and sustained return behavior, rather than only maximizing authentication events.

Three principles guide this strategy: First, prioritize player outcomes. The product team should measure how many players return, not only how frequently existing active players authenticate. Second, evaluate long term effects. Short term reward interaction or increased early return activity should not be considered sufficient evidence of sustained engagement improvement. Third, protect commercial performance. The monetization analysis does not demonstrate a consistently positive relationship between authentication frequency and ARPU. Any engagement intervention must be evaluated against revenue and reward economy guardrails.

9.2 Implementation Roadmap

Phase 1 - Discovery & Specification

Finalize feature eligibility, reward progression rules, technical requirements, economy constraints and experiment success criteria.

Phase 2 - Prototype & Instrumentation

Build the initial feature, implement event tracking and verify assignment integrity, event completeness and KPI calculations.

Phase 3 - Controlled Experiment

Randomize eligible players into control and treatment groups. Monitor implementation quality and guardrails, then allow complete observation of the primary retention outcome.

Phase 4 - Evaluation & Rollout

Evaluate effect size, confidence intervals, practical significance, and guardrails. Decide whether to rollout, iterate or discontinue.

9.3 Product Management Conclusion

This project demonstrates a structured approach to identifying, prioritizing and validating a product improvement opportunity using behavioral and monetization data. The findings support a product strategy focused on increasing sustained return behavior while protecting commercial outcomes.

Progressive Return Rewards is selected as the preferred candidate based on the RICE assessment. Its effectiveness remains unproven and must be established through controlled experimentation.

The final recommendation is to prioritize the sustained engagement problem, validate the proposed intervention through a properly designed experiment, and make the rollout decision based on retention impact, statistical evidence and business guardrails.

This approach connects descriptive analytics with product strategy, feature prioritization and measurable decision making without overstating what the available data can establish.